Chapter 7
Analytical Methods for the Propagation of Uncertainties



7.1 Introduction

This chapter presents analytical methods for propagating uncertainties through an indirect measurement model. The method was first introduced by Carl Gauss in the 1820s (Gauss 1821), and was further developed by Kline and McClintock (1953). In 1995, the Joint Committee for Guides in Metrology standardized the method using it as the basis for its GUM (JCGM 100 2008), so the technique is often referred to as the GUM method for uncertainty propagation.

Given an indirect measurement model, the measurand is a function of multiple measured values:

$$Y = f(X_1, X_2, \ldots, X_N), \tag{7.1}$$

where the inputs X_1, X_2, …, X_N are the physical quantities to be measured and the function f relates the measurands to the output quantity Y. As an example, the output quantity Y could represent the density of a cylinder, while X_1, X_2, and X_3 are quantities of mass, diameter, and height, and f is the functional relationship between the quantities (density = mass/volume).

An estimate of Y, denoted y, is determined by making measurements—or estimates—of each quantity. These estimates are denoted $x_1, x_2, \ldots, x_N$ (see GUM Sect. 4.1.4). Then y is computed using $y = f(x_1, x_2, \ldots x_n)$. The input estimates are often the average of multiple independent measurements on the same quantity. Each input estimate has an associated standard uncertainty estimate $u(x_i)$, computed from Type A and Type B uncertainty components as described in Chap. 6. Continuing the density example, an estimate of cylinder density is $y = \rho = f(m, d, h) = \frac{4m}{\pi h d^2}$, where m, d, and h are the average measured values of mass, diameter, and height, respectively.

The remainder of this section considers how to determine $u_c(y)$, the combined standard uncertainty in the output estimate y. For a small subset of problems, $u_c(y)$

© Springer Nature Switzerland AG 2020

S. Crowder et al., *Introduction to Statistics in Metrology*,
https://doi.org/10.1007/978-3-030-53329-8_7

can be calculated exactly, but in most cases, the method makes an approximation to linearize the measurement function. A derivation of the uncertainty expression introduced in Chap. 6 (Eq. 6.34) assumes the measurement model can be linearly approximated and the inputs are uncorrelated. The correlated terms are then retained in the uncertainty derivation. Finally, the case where nonlinear terms in the measurement model become significant is examined.

7.2 Mathematical Basis

This section reviews several important properties of the sums of random variables. These results will be used, along with a Taylor series approximation of Eq. (7.1), to estimate the combined uncertainty of any arbitrary measurement equation.

First, define $E(\cdot)$ as the expectation operator, $\mathrm{Var}(\cdot)$ as the variance operator, and $\mathrm{Cov}(\cdot,\cdot)$ as the covariance operator for random variables (see Casella and Berger 2002). The expectation is simply the weighted average of a probability distribution, the variance $\mathrm{Var}(X) = E(X - E(X))^2$ is a measure of spread of the distribution, equal to the square of the standard deviation, and the covariance is a measure of the strength of the relationship between two random variables. If $X_1, X_2, \ldots, X_N$ are random variables (see GUM C.2.2) and $a_1, a_2, \ldots, a_N$ are constants, then some useful properties of expectation and variance are:

$$E(X + a) = E(X) + a$$

$$E(aX) = aE(X)$$

$$E(X_1 + X_2) = E(X_1) + E(X_2)$$

$$\mathrm{Var}(X + a) = \mathrm{Var}(X)$$

$$\mathrm{Var}(aX) = a^2\mathrm{Var}(X)$$

$$\mathrm{Var}(X_1 + X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + 2\mathrm{Cov}(X_1, X_2)$$

The above properties can be combined and generalized to multiple random variables:

$$E\left(\sum_{i=1}^{N} a_i X_i\right) = \sum_{i=1}^{N} a_i E(X_i),$$

and

$$\mathrm{Var}\left(\sum_{i=1}^{N} a_i X_i\right) = \sum_{i=1}^{N} a_i^2 \mathrm{Var}(X_i) + 2\sum_{i=1}^{N-1}\sum_{j=i+1}^{N} a_i a_j \mathrm{Cov}\left(X_i, X_j\right) \tag{7.2}$$

If the $X_i's$ are uncorrelated, the covariance terms are 0, and Eq. (7.2) reduces to

$$\mathrm{Var}\left(\sum_{i=1}^{N} a_i X_i\right) = \sum_{i=1}^{N} a_i^2 \mathrm{Var}(X_i) \tag{7.3}$$

To determine the expectation and variance of an indirect measurand Y, one must find the expected value of Y, $E(Y) = E[f(X_1, X_2, \ldots, X_N)]$, and the variance of Y, $\mathrm{Var}(Y) = \mathrm{Var}[f(X_1, X_2, \ldots, X_N)]$. For example, consider a model that converts measurements units such as meters to centimeters: $f(X) = aX$. Using 7.3 the variance of $f(X)$ is $a^2\mathrm{Var}(X) = a^2u^2(x)$. Therefore, the uncertainty of the converted value is the uncertainty of the measurement of x in original units multiplied by the units conversion factor a.

Equations (7.2) and (7.3) allow for finding the expectation and variance of a measurement model consisting of a linear combination of variables, such as simple shifting ($f(X) = X + a$), scaling ($f(X) = aX$), or adding multiple scaled variables ($f(X_1, X_2) = a_1X_1 + a_2X_2$). In these cases, an exact variance is computed. Of course, most measurement models do not fall into one of these simple linear combinations, so approximation of more complex measurement equations must be considered. The simplest approximate case uses a linear approximation of the nonlinear model with uncorrelated variables, considered in Sect. 7.3.

7.3 The Simple Case: First-Order Terms with Uncorrelated Inputs

The previous section considered propagating uncertainty for linear combinations of multiple measurements. Most measurement models, however, will be nonlinear. The approach for an arbitrary (nonlinear) measurement equation $Y = f(X_1, X_2)$ is to approximate the equation with a first-order (linear) Taylor series expansion. The derivation of the estimated combined standard uncertainty $u_c(y)$ uses $u^2(x_i)$ to represent an estimate of $Var(X_i)$ and $u(x_i, x_j)$ to represent an estimate of $\mathrm{Cov}(X_i, X_j)$. The "realization" or "estimate" of the random variable X_i is denoted by x_i,and the estimate of the measurand Y is denoted by y.

Expanding about the input variables' respective means, μ_{X_1} and μ_{X_2}, the first-order approximation takes the form:

where $k = t_p(\nu_{\text{eff}})$ is computed from the t-distribution with confidence level p and effective degrees of freedom ν_{eff}. The Welch–Satterthwaite approximation can be used to calculate ν_{eff}:

$$\nu_{\text{eff}} = \frac{u_c^4(y)}{\sum_{i=1}^{N} \frac{(c_i u(x_i))^4}{\nu_i}} \tag{7.9}$$

Here, the c_i variables are the "sensitivity coefficients," which are simply shorthand for the partial derivative terms computed previously: $c_i = \frac{\partial f}{\partial x_i}$.

7.3.1 Measurement Examples

Consider measuring the density of a cylinder. The mass (m), diameter (d), and height (h) are measured, and the density found from the measurement equation:

$$\rho = \frac{m}{\pi \left(\frac{d}{2}\right)^2 h} \tag{7.10}$$

```
# Measurement equation
rho <- function(m, h, d) {m/(pi*(d/2)^2*h)}
```

The measured m, h, and d values, and their standard uncertainties $u(m)$, $u(h)$, and $u(d)$ are all known from the measurement results and setup (Table 7.1).

```
# Mean values and uncertainties for variables
m <- c(5.0, 0.001)
h <- c(2.0, 0.005)
d <- c(0.5, 0.005)
```

Applying Eq. (7.7), the expression for the combined variance in density is

$$u_c^2(\rho) = \left(\frac{\partial \rho}{\partial m}\right)^2 u^2(m) + \left(\frac{\partial \rho}{\partial h}\right)^2 u^2(h) + \left(\frac{\partial \rho}{\partial d}\right)^2 u^2(d) \tag{7.11}$$

The sensitivity coefficients (denoted as c_x) are:

Table 7.1 Measured values and uncertainties of cylinder density determination

Variable	Mean	Uncertainty
m	5.0 g	0.001 g
h	2.0 cm	0.005 cm
d	0.5 cm	0.005 cm

If all the measurement uncertainties have infinite degrees of freedom, $k = 1.96$ gives a 95% confidence interval, and the density is

$$\rho = 12.73\text{g/cm}^3 \pm 0.52\text{g/cm}^3 \ (95\%, k = 1.96). \tag{7.15}$$

```
#95% Confidence interval
k <- qnorm((1 - (.05/2)))
confInt <- k * uComb
confInt
## [1] 0.5030096
```

To identify the variable that is the biggest contributor to total uncertainty, we compute the proportion of the total variance that comes from each variable via the formula: $\frac{c_x^2 u^2(x)}{u_c^2(y)} \times 100\%$.

For this example, $u(m)$ contributes 0.01%, $u(h)$ contributes 1.54%, and $u(d)$ contributes 98.45% of the total uncertainty. If the goal is to reduce the total uncertainty, the focus should be on improving the uncertainty in the diameter measurement.

```
# Proportion of Variance
umProp <- um^2/uComb^2
umProp
## [1] 9.845184e-05

uhProp <- uh^2/uComb^2
uhProp
## [1] 0.0153831

udProp <- ud^2/uComb^2
udProp
## [1] 0.9845184
```

7.4 First-Order Terms with Correlated Inputs

In the previous derivation, the covariance term was set to 0 because the input variables were assumed to be uncorrelated. This is often not the case, so a complete derivation beginning with the Taylor series and retaining the covariance term is given here. Taking the same approach as the simple models in Sect. 7.3, f is approximated by the first-order Taylor series:

$$Y = f(X_1, X_2) \approx f\left(\mu_{X_1}, \mu_{X_2}\right) + \frac{\partial f}{\partial X_1}\left(X_1 - \mu_{X_1}\right) + \frac{\partial f}{\partial X_2}\left(X_2 - \mu_{X_2}\right). \tag{7.16}$$

Applying Eq. (7.2) term by term (including the covariance term), the combined variance is

$$\begin{aligned} \text{Var}\,(Y) &\approx \text{Var}\left(f\left(\mu_{X_1}, \mu_{X_2}\right) + \frac{\partial f}{\partial X_1}\left(X_1 - \mu_{X_1}\right) + \frac{\partial f}{\partial X_2}\left(X_2 - \mu_{X_2}\right)\right) \\ &= \text{Var}\left(\frac{\partial f}{\partial X_1}\left(X_1 - \mu_{X_1}\right)\right) + \text{Var}\left(\frac{\partial f}{\partial X_2}\left(X_2 - \mu_{X_2}\right)\right) \\ &\quad +2\text{Cov}\left(\frac{\partial f}{\partial X_1}\left(X_1 - \mu_{X_1}\right), \frac{\partial f}{\partial X_2}\left(X_2 - \mu_{X_2}\right)\right), \end{aligned} \tag{7.17}$$

with all other variance and covariance terms equal to 0. This expression yields the following estimated combined variance, recalling that $u(x_1, x_2)$ represents an estimate of the covariance of X_1 and X_2:

$$u_c^2(y) = \left(\frac{\partial f}{\partial x_1}\right)^2 u^2(x_1) + \left(\frac{\partial f}{\partial x_2}\right)^2 u^2(x_2) + 2\frac{\partial f}{\partial x_1}\frac{\partial f}{\partial x_2} u(x_1, x_2). \tag{7.18}$$

This result can also be generalized to any number of input variables as

$$u_c^2(y) = \sum_{i=1}^{N}\left(\frac{\partial f}{\partial x_i}\right)^2 u^2(x_i) + 2\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\frac{\partial f}{\partial x_i}\frac{\partial f}{\partial x_j} u\left(x_i, x_j\right). \tag{7.19}$$

Note that when the covariance for all combinations of inputs is zero, this reduces to the expression found in Sect. 7.2 (Eq. 7.7).

7.4.1 Covariance, Correlation, and Effect on Uncertainty

It is often easier to work in terms of a correlation coefficient (Eq. 4.4) than a covariance. The correlation coefficient always ranges from −1 to +1, with uncorrelated values having a coefficient of 0. To convert covariance into correlation $r(x_i, x_j)$, use

$$r\left(x_i, x_j\right) = \frac{u\left(x_i, x_j\right)}{u(x_i)u\left(x_j\right)}. \tag{7.20}$$

Some common cases of correlated inputs occur when two quantities in the model are measured using the same piece of equipment, when temperature affects two or

more measurements assuming the measurements are made at the same temperature, or when multiple measurements are made simultaneously (as in the resistance example in Sect. 7.4.2). In many cases, the correlation is insignificant, but it can also be difficult to quantify. One technique is to take multiple measurements of the same sample and estimate the correlation between each input using the sampled data (Eq. 4.4). In some cases, for example, with temperature correlations, it may be easier to include temperature as an independent variable in the model itself rather than trying to determine a correlation coefficient between the other variables.

Correlated variables do not necessarily make the total uncertainty higher or lower. The direction of change due to correlation depends entirely on the sign of the sensitivity coefficient and the sign of the correlation coefficient. Consider measurements of electrical resistance using the model $R = V/I$. Applying Eq. (7.18) to this expression and computing the partial derivatives, it is easy to show that the constant multiplier in the covariance term of the uncertainty expression is $-(2V/I^3)$. Voltage and current always have the same sign in this situation, so a positive covariance always *decreases* the total uncertainty. However, if the same measurements were used to determine electrical power using the model $P = IV$, the multiplier in the covariance term is $2IV$,and a positive covariance *increases* the total uncertainty.

7.4.2 Measurement Examples

Semiconductor nanowires are grown by a Vapor–Liquid–Solid method in a furnace under tightly controlled conditions of temperature, pressure, and atmospheric composition. To use a very simplified model of the complex chemistry of nanowire growth, material properties of the nanowires, such as electrical resistivity, depend on the furnace conditions, while the length and diameter dimensions depend on the amount of time spent in the furnace. Because both length and diameter depend on time, these variables are correlated.

The expected resistance of a batch of nanowires can be calculated using the resistivity (ρ), length (L), and diameter (d):

$$R = \frac{4\rho L}{d^2} \tag{7.21}$$

```
# Measurement equation
R <- function(rho, L, d) {(4*rho*L)/d^2}
```

The furnace has been well characterized, so for a given process the resistivity is known to be 9.6×10^{-5} Ω-m with uncertainty of 2.0×10^{-6} Ω-m. After growing a batch of wires using this process, 20 wires were sampled and the dimensions measured to be $d = 180$ nm $\pm$ 20 nm and $L = 5.2$ μm $\pm$ 0.20 μm (all $k = 1$ standard

which gives a 95% confidence k-factor of 2.16. Therefore, the expected resistance for this batch of nanowires, to 95% confidence, reported to the typical 2 significant figures in uncertainty, is

$$R = 62\ \text{k}\Omega \pm 27\ \text{k}\Omega\ (95\%, k = 2.16). \tag{7.26}$$

```
# Value of R
Rval <- R(rho = rho[1], L = L[1], d = d[1])
Rval
## [1] 61629.63
#95% Confidence interval
k <- qt((1 - (.05/2)), 13)
confInt <- k * uCombCorr
confInt
## [1] 26746.73
```

7.5 Higher-Order Terms with Uncorrelated Inputs

The analytical method discussed so far uses a first-order Taylor series approximation of the measurement function. This assumes that the function can be modelled linearly in the region of interest (near the mean values of the measurands).

What does it mean to be nonlinear in this sense? Consider the model $f(x) = e^x$ shown in Fig. 7.1. On the left side, a small x uncertainty is given. Over a small range like this, the exponential can be treated as linear without introducing significant error. On the right side, x has a much larger uncertainty, and a linear approximation is no longer sufficient to cover the entire range of possible $f(x)$ values. Any function can be approximated as linear if the uncertainties are small enough. Unfortunately, it is difficult to know beforehand whether this is true for a particular case without calculating the nonlinear effects and comparing to the required number of significant digits.

To include nonlinear effects in the uncertainty expression, a higher-order Taylor series can be used. Here the second-order terms of the approximation are included for two input variables:

$$\begin{aligned} Y = f(X_1, X_2) \approx{}& f\left(\mu_{X_1}, \mu_{X_2}\right) + \frac{\partial f}{\partial X_1}\left(X_1 - \mu_{X_1}\right) + \frac{\partial f}{\partial X_2}\left(X_2 - \mu_{X_2}\right) \\ &+ \frac{1}{2}\frac{\partial^2 f}{\partial X_1^2}\left(X_1 - \mu_{X_1}\right)^2 + \frac{\partial f}{\partial X_1}\frac{\partial f}{\partial X_2}\left(X_1 - \mu_{X_1}\right)\left(X_2 - \mu_{X_2}\right) \\ &+ \frac{1}{2}\frac{\partial^2 f}{\partial X_2^2}\left(X_2 - \mu_{X_2}\right)^2 \end{aligned} \tag{7.27}$$

$$u_c^2(y) = \sum_{i=1}^{N} \left(\frac{\partial f}{\partial x_i}\right)^2 u^2(x_i) + \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)^2 u^2(x_i)u^2(x_j) + \sum_{i=1}^{N}\frac{\partial f}{\partial x_i}\frac{\partial^3 f}{\partial x_i^3}u^4(x_i) + \sum_{i=1}^{N}\sum_{\substack{j=1 \\ j\neq i}}^{N}\frac{\partial^2 f}{\partial x_i^2}\left(\frac{\partial f}{\partial x_j}\right)^2 u^2(x_i)\,u^2(x_j) + \frac{1}{4}\left(\sum_{i=1}^{N}\frac{\partial^2 f}{\partial x_i^2}u^2(x_i)\right)^2 \tag{7.29}$$

Because of the complexity of these formulas, the need for higher-order partial derivatives, and the ignoring of correlated variables, it is often easier to use numerical approaches such as Monte Carlo methods described in Chap. 8 to account for nonlinearities in the measurement model.

7.5.1 Measurement Examples

Returning to the density measurement example, the same cylinder was measured using different equipment. Because of the larger uncertainties in this equipment, it is suspected that the linearity assumption may no longer be valid (Table 7.2).

$$\rho = \frac{m}{\pi\left(\frac{d}{2}\right)^2 h} \tag{7.30}$$

Following the same calculation as in Sect. 7.3.1, using the first-order uncertainty expression, and assuming all variables are independent, the density ($k = 1$) is

$$\rho = 12.7\text{g/cm}^3 \pm 2.57\text{g/cm}^3 \tag{7.31}$$

Applying the GUM's expression of nonlinear uncertainty (Eq. 7.28) adds in the higher-order terms:

Table 7.2 Cylinder density measurement values using different equipment

Variable	Mean	Uncertainty
m	5.0 g	0.01 g
h	2.0 cm	0.05 cm
d	0.5 cm	0.05 cm

$$
\begin{aligned}
u_{c,2}^2(\rho) = \frac{16}{\pi^2 d^4 h^2} u^2(m) + \frac{16m^2}{\pi^2 d^4 h^4} u^2(h) + \frac{64m^2}{\pi^2 d^6 h^2} u^2(d) \\
+ \frac{16}{\pi^2 d^4 h^4} u^2(h) u^2(m) \\
+ \frac{32m^2}{\pi^2 d^4 h^6} u^4(h) + \frac{64}{\pi^2 d^6 h^2} u^2(d) u^2(m) \\
+ \frac{64m^2}{\pi^2 d^6 h^4} u^2(d) u^2(h) + \frac{288m^2}{\pi^2 d^8 h^2} u^4(d) \\
+ \frac{96m}{\pi d^2 h^5} u^4(h) + \frac{96m}{\pi d^4 h^3} u^2(d) u^2(h) + \frac{480m}{\pi d^6 h} u^4(d)
\end{aligned}
\tag{7.32}
$$

which simplifies to a density of

$$\rho = 12.7\text{g/cm}^3 \pm 2.65\text{g/cm}^3 \tag{7.33}$$

Note that this result is only a slightly different uncertainty than using the first-order approximation, but it is a slightly better approximation to the true uncertainty. Using the even more complete expression in Wang and Iyer (2005) (Eq. 7.29) results in a density of

$$\rho = 12.7\text{g/cm}^3 \pm 2.72\text{g/cm}^3. \tag{7.34}$$

In theory, one could keep adding terms to the Taylor expansion until the resulting combined uncertainty did not change within the reported significant digits. If the above results were rounded and reported to the typical two significant digits, adding the additional terms of the Wang and Iyer method would not change the result compared to the GUM method.

7.6 Multiple Output Quantities

Occasionally, a measurement model will result in multiple output quantities based on the same set of input measurements. One common example is encountered in microwave measurements, where the real and imaginary components of a complex-valued microwave signal are input to a measurement model that results in magnitude and phase as outputs. Other examples include measurement of voltage and current used to calculate both resistance and power, or a set of Coordinate Measuring Machine (CMM) measurements used to compute both the lengths and angle of intersection of two components.

Cases with multiple output quantities, dealt with extensively in JCGM 102 (2011), result in an "uncertainty region" in multiple dimensions, rather than the simple one-dimensional "plus or minus" uncertainty region evaluated previously

in this chapter. Part of a full solution of a multiple output uncertainty evaluation is determining the joint probability between the multiple outputs in order to fully define the region of uncertainty.

In the bivariate case, the joint PDF of the resulting uncertainty calculation can be described by a bivariate normal distribution that defines an ellipse. A bivariate normal PDF requires the mean values of the two output quantities y_1 and y_2 and their covariance matrix U_y. Defining U_x as the covariance matrix of the input measurements, the sensitivity matrix

$$C_x = \begin{bmatrix} \frac{\partial f_1}{\partial X_1} & \frac{\partial f_1}{\partial X_2} \\ \frac{\partial f_2}{\partial X_1} & \frac{\partial f_2}{\partial X_2} \end{bmatrix} \tag{7.35}$$

with the partial derivatives evaluated at the measured values $X_i = x_i$, the covariance matrix of the two output quantities is given by

$$U_y = C_x U_x C_x^T = \begin{bmatrix} u^2(y_1) & u(y_1, y_2) \\ u(y_1, y_2) & u^2(y_2) \end{bmatrix} \tag{7.36}$$

These matrices can be expanded to any number of output quantities.

7.7 Limitations of the Analytical Approach

The analytical approach to uncertainty propagation has its limitations. Except for a few very simple cases (those described in Sect. 7.2), the solution is an approximation based on a Taylor series expansion. It is not obvious when nonlinearities in the model will become significant, requiring use of higher-order terms. The solutions all require finding partial derivatives of the measurement model, which in some cases can be difficult or impossible to solve analytically.

Another limitation is the normality assumption. All input variables are defined by their mean and standard deviation, regardless of their actual distribution. Any information about a distribution (e.g., uniform or triangular) is lost after converting to a standard deviation as described in Sect. 6.3. As such, the distribution of the function's output will always be treated as a normal distribution, defined by a mean and standard deviation.

Finally, using the W-S formula to determine effective degrees of freedom and an expanded uncertainty is only justified when the model is first-order with uncorrelated inputs. For these reasons, the Monte Carlo method described in Chap. 8 is often used in parallel with this section's analytical methods to help confirm the results.

7.8 Related Reading

The paper by Kline and McClintock (1953), and the follow-up paper by Kline (1985), on which the uncertainty propagation methods described in this chapter are based, is recommended reading simply for historical reasons. NIST published Technical Note 1297 (Taylor and Kuyatt 1994) which provides additional guidance on using the GUM method and expressing the results. NIST Technical Note 1900 (Possolo 2015) provides many practical examples demonstrating use of the GUM uncertainty methods.

The GUM (JCGM 2008), also released as ISO/IEC Guide 98-3:2008, is the industry standard for guidance on uncertainty propagation, covering many additional aspects of uncertainty analysis. ISO 21748:2017 provides guidance for evaluating uncertainty due to repeatability and reproducibility in experimental measurements. NCSLI's recommended practice RP-12 National Conference of Calibration and Standards Laboratories International (1995) provides additional recommendations on how to practically implement uncertainty calculations. Suggestions include methods of identifying sources of uncertainty in a measurement, selecting appropriate probability distributions, and choosing a coverage factor.

Hall and Willink (2001) investigate the Welch–Satterthwaite approximation and when it does or does not apply. Work on adapting the W-S approximation for correlated uncertainties is ongoing in Castrup (2010), Wang and Iyer (2005), Ballico (2000), Willink (2007), and Kuster (2013). Some other shortcomings of the GUM uncertainty method, such as additional correlation effects and nonlinearities, are explored in detail in Rabinovich (2005).

7.9 Exercises

1. Use the GUM uncertainty formula (Eq. 7.7) to find an expression for the standard uncertainty in:

   (a) $f = x$
   (b) $f = x + y$
   (c) $f = x + y + z$

   Given uncertainties $u(x)$, $u(y)$, and $u(z)$ and assuming x, y, and z are independent (uncorrelated). Write a general expression for uncertainty in $f = \sum x_i$ with uncertainties $u(x_i)$.
2. Use the GUM uncertainty formula (Eq. 7.7) to compute the standard uncertainty in $f(x) = cx$, where c is a constant with no uncertainty. Apply the result to find the uncertainty in kilograms of a mass is given as 355 g with uncertainty 5.2 g.
3. Find an expression for the uncertainty in $f(x, y) = x^2 + y$. Then solve for:

   (a) $x = 4$, $y = 3$, $u(x) = 0.4$, $u(y) = 0.15$,
   (b) $x = 4$, $y = 3$, $u(x) = 10\%$, $u(y) = 5\%$.

References

Ballico, M.: Limitations of the Welch-Satterthwaite approximation for measurement uncertainty calculations. Metrologia. **37**(1), 61–64 (2000)
Casella, G., Berger, R.L.: Statistical Inference, 2nd edn. Duxbury, Pacific Grove, CA (2002)
Castrup, H.: A Welch-Satterthwaite relation for correlated errors. In: Measurement Science Conference, Pasadena, CA (2010)
Gauss, C.F.: Theoria combinationis observationum erroribus minimis obnoxiae. Carl Friedrich Gauss Werke. **4**, 95–100 (1821)
Hall, B.D., Willink, R.: Does Welch-Satterthwaite make a good uncertainty estimate? Metrologia. **38**, 9–15 (2001)
ISO 21748: Guidance for Use of Repeatability, Reproducibility, and Trueness Estimates in Measurement Uncertainty and Evaluation. (2017)
JCGM 100: Evaluation of Measurement Data - Guide to the Expression of Uncertainty in Measurement. (2008)
JCGM 102: Evaluation of Measurement Data - Supplement 2 to the - Guide to the Expression of Uncertainty in Measurement - Extension to any Number of Output Quantities. (2011)
Kline, S. J., McClintock, F. A.: Describing uncertainties in single sample experiments. Mech. Eng., 75 3–8 (1953)
Kline, S.J.: The purposes of uncertainty analysis. J. Fluids Eng. **107**, 153–160 (1985)
Kuster, M.: Applying the Welch-Satterthwaite formula to correlated errors. NCSLI Meas. **8**(1), 42–55 (2013)
National Conference of Calibration and Standards Laboratories International: Recommended Practice 12 - Determining and Reporting Measurement Uncertainties. (1995)
Possolo, A.: Simple guide for evaluating and expressing the uncertainty of NIST measurement results. https://doi.org/10.6028/NIST.TN.1900 (2015). Accessed 29 Apr 2020
Rabinovich, S.G.: Measurement Errors and Uncertainties, 3rd edn. Springer, New York, NY (2005)
Taylor, B.N., Kuyatt, C.E.: Guidelines for evaluating and expressing the uncertainty of NIST measurement results. NIST Technical Note 1297. National Institute of Standards and Technology, Gaithersburg, MD (1994)
Time and Frequency from A to Z. https://www.nist.gov/pml/time-and-frequency-division/popular-links/time-frequency-z (2010). Accessed 29 Apr 2020
Walkup, N.: Eratosthenes and the mystery of the Stades. Mathematical Association of America. https://www.maa.org/press/periodicals/convergence/eratosthenes-and-the-mystery-of-the-stades-introduction (2005). Accessed 29 Apr 2020
Wang, C.M., Iyer, H.K.: On higher-order corrections for propagating uncertainties. Metrologia. **42**, 406–410 (2005)
Willink, R.: A generalization of the Welch-Satterthwaite formula for use with correlated uncertainty components. Metrologia. **44**, 340–349 (2007)
Zuckerwar, A.J.: Handbook of the Speed of Sound in Real Gasses. Academic Press, San Diego, CA (2002)